

Use-Case Flow: UC-02 Submit Elective Bid

Academic Elective Bidding & Allocation System — Problem Statement #05

Primary Actor	Student
Secondary Actor(s)	Allocation System (prerequisite / credit validation engine)
Includes	UC-01 Authenticate User, UC-03 Validate Prerequisites
Trigger	The elective bidding window opens and the Student chooses to submit or edit a bid.

Preconditions

1. The bidding window for the current term is open.
2. The Student is a registered, active student with a valid account.
3. The Academic Registrar has published the elective catalog (courses, seats, slots, prerequisites) for the term.

Postconditions

Success: The bid — ranked preferences (3–8 electives) with credits totaling exactly 100 — is persisted, timestamped, and marked "Submitted"; all ranked electives pass prerequisite validation.

Failure: No bid record is created (or an existing bid is unchanged); the Student sees the specific validation error and may retry before the window closes.

Main Success Scenario

1. Student logs in to the bidding portal (include: UC-01 Authenticate User).
2. System displays the published elective catalog: course codes, seats, timetable slots, and prerequisites.
3. Student selects 3–8 electives and ranks them in strict order (1 = most preferred).
4. Student allocates a portion of 100 bidding credits to each ranked elective so the total equals exactly 100.
5. Student submits the bid.
6. System validates prerequisite completion for every ranked elective (include: UC-03 Validate Prerequisites).
7. System validates that total credits equal exactly 100 and no elective is ranked twice.
8. System persists the bid, timestamps it, and marks it "Submitted."
9. System displays a confirmation summarizing ranked preferences and credit allocation; states the bid remains editable until the window closes.
10. Use case ends successfully.

Alternate Flow — A1: Prerequisite Validation Fails

Branches from step 6 of the Main Success Scenario.

6a1. System finds that one or more ranked electives has an unmet prerequisite.

6a2. System rejects the submission without persisting the bid, and displays an error identifying each affected elective and its missing prerequisite.

6a3. System returns the Student to the bid-editing screen with in-progress selections preserved.

6a4. Student removes the ineligible elective(s), redistributes credits, and resumes at step 4 of the Main Success Scenario — or submits a manual prerequisite waiver request to the Academic Registrar (handled outside this use case) and returns later to resubmit.